

should consider addressing both immediate displacement and longer-term structural changes through, for example, education reform and alternative skill development pathways.

Several important limitations qualify our findings. First, our identification strategy cannot rule out other shocks that differentially affected high- versus low-exposure firms or occupations, such as sector-specific demand shifts, regulatory changes, or complementary technological developments that correlate with LLM susceptibility, although our robustness checks provide evidence against most of these channels. Second, the LinkedIn-based employment data likely overrepresents professional and technical workers who maintain online profiles while underrepresenting manual, service, and informal sector employment, potentially inflating our estimates if LLM impacts concentrate among digitally-active occupations. Third, our exposure measures from Eloundou et al. (2024) capture potential vulnerability to LLM disruption rather than actual adoption, which depends on unobserved factors including financial resources, management quality, technological infrastructure, and organizational culture. Firms with high exposure scores may resist adoption while nominally less-exposed firms might implement LLMs aggressively, attenuating our reduced-form estimates. Finally, our framework cannot observe the creation of entirely new occupational categories enabled by LLM technology such as prompt engineers, AI safety specialists, or human-AI collaboration designers, which historical evidence suggests constitutes a primary channel through which technological change generates employment (Restrepo, 2024).

These limitations highlight the preliminary nature of our findings and provide potential avenues for future research. Most fundamentally, our 30-month observation window may capture only short-term labor market disruption rather than long-term equilibrium outcomes. Previous general-purpose technologies required decades for complementary innovations, business process reengineering, and workforce retraining to fully work their way through the labor market. The initial displacement we document might reverse as organizations develop new LLM-augmented tasks where humans maintain comparative advantage, or intensify as models become more capable and adoption barriers fall. In fact, we already see some evidence that the decrease in hiring in highly exposed occupations attenuates towards the end of the sample period. Future research should extend the analysis period to distinguish between temporary adjustment and permanent structural change. Additionally, future

research could examine heterogeneity in adoption patterns and employment effects across firm characteristics, investigate whether LLM exposure affects job quality and working conditions, and examine a wider set of countries.

References

- Acemoglu, D. and Restrepo, P. (2018), ‘The race between man and machine: Implications of technology for growth, factor shares, and employment’, *American Economic Review* **108**(6), 1488–1542.
- Acemoglu, D. and Restrepo, P. (2019), ‘Automation and new tasks: How technology displaces and reinstates labor’, *Journal of Economic Perspectives* **33**(2), 3–30.
- Acemoglu, D. and Restrepo, P. (2020), ‘Robots and jobs: Evidence from US labor markets’, *Journal of Political Economy* **128**(6), 2188–2244.
- Autor, D. (2024), Applying AI to rebuild middle class jobs, Technical report, National Bureau of Economic Research.
- Autor, D. H. (2015), ‘Why are there still so many jobs? the history and future of workplace automation’, *Journal of Economic Perspectives* **29**(3), 3–30.
- Autor, D. H., Levy, F. and Murnane, R. J. (2003), ‘The skill content of recent technological change: An empirical exploration’, *The Quarterly Journal of Economics* **118**(4), 1279–1333.
- Autor, D. and Thompson, N. (2025), ‘Expertise’, *Journal of the European Economic Association* p. jvaf023.
- Brynjolfsson, E., Chandar, B. and Chen, R. (2025), ‘Canaries in the coal mine? six facts about the recent employment effects of artificial intelligence’. Working paper.
- Brynjolfsson, E., Li, D. and Raymond, L. (2025), ‘Generative AI at work’, *The Quarterly Journal of Economics* **140**(2), 889–942.
- Chandar, B. (2025), ‘Tracking employment changes in ai-exposed jobs’. Working paper, SSRN.
- del Rio-Chanona, R. M., Ernst, E., Merola, R., Samaan, D. and Teutloff, O. (2025), ‘AI and jobs: A review of theory, estimates, and evidence’, *arXiv preprint arXiv:2509.15265*.

- Dominski, J. and Lee, Y. S. (2025), ‘Advancing ai capabilities and evolving labor outcomes’. Working paper, arXiv:2507.08244.
- Eisfeldt, A. L., Schubert, G. and Zhang, M. B. (2023), Generative AI and firm values, Technical report, National Bureau of Economic Research.
- Eloundou, T., Manning, S., Mishkin, P. and Rock, D. (2024), ‘GPTs are GPTs: Labor market impact potential of LLMs’, *Science* **384**(6702), 1306–1308.
- Fadhel, A., Panella, K., Rouen, E. and Serafeim, G. (2021), ‘Accounting for employment impact at scale’, *Harvard Business School Accounting & Management Unit Working Paper* (22-018).
- Hampole, M., Papanikolaou, D., Schmidt, L. D. and Seegmiller, B. (2025), Artificial intelligence and the labor market, Technical report, National Bureau of Economic Research.
- Henseke, G., Davies, R., Felstead, A., Gallie, D., Green, F. and Zhou, Y. (2025), ‘How exposed are UK jobs to generative AI? Developing and applying a novel task-based index’, *arXiv preprint arXiv:2507.22748*.
- Humlum, A. and Vestergaard, E. (2025), Large language models, small labor market effects, Technical report, National Bureau of Economic Research.
- Ide, E. (2025), ‘Automation, ai, and the intergenerational transmission of knowledge’, *arXiv preprint arXiv:2507.16078*.
- Lichtinger, G. and Hosseini Maasoum, S. M. (2025), ‘Generative AI as seniority-biased technological change: Evidence from US résumé and job posting data’, *Available at SSRN*.
- Noy, S. and Zhang, W. (2023), ‘Experimental evidence on the productivity effects of generative artificial intelligence’, *Science* **381**(6654), 187–192.
- Office for National Statistics (2022), ‘Employment in the uk: September 2022’. Accessed: 17-09-2025.
- Restrepo, P. (2024), ‘Automation: Theory, evidence, and outlook’, *Annual Review of Economics* **16**(Volume 16, 2024), 1–25.